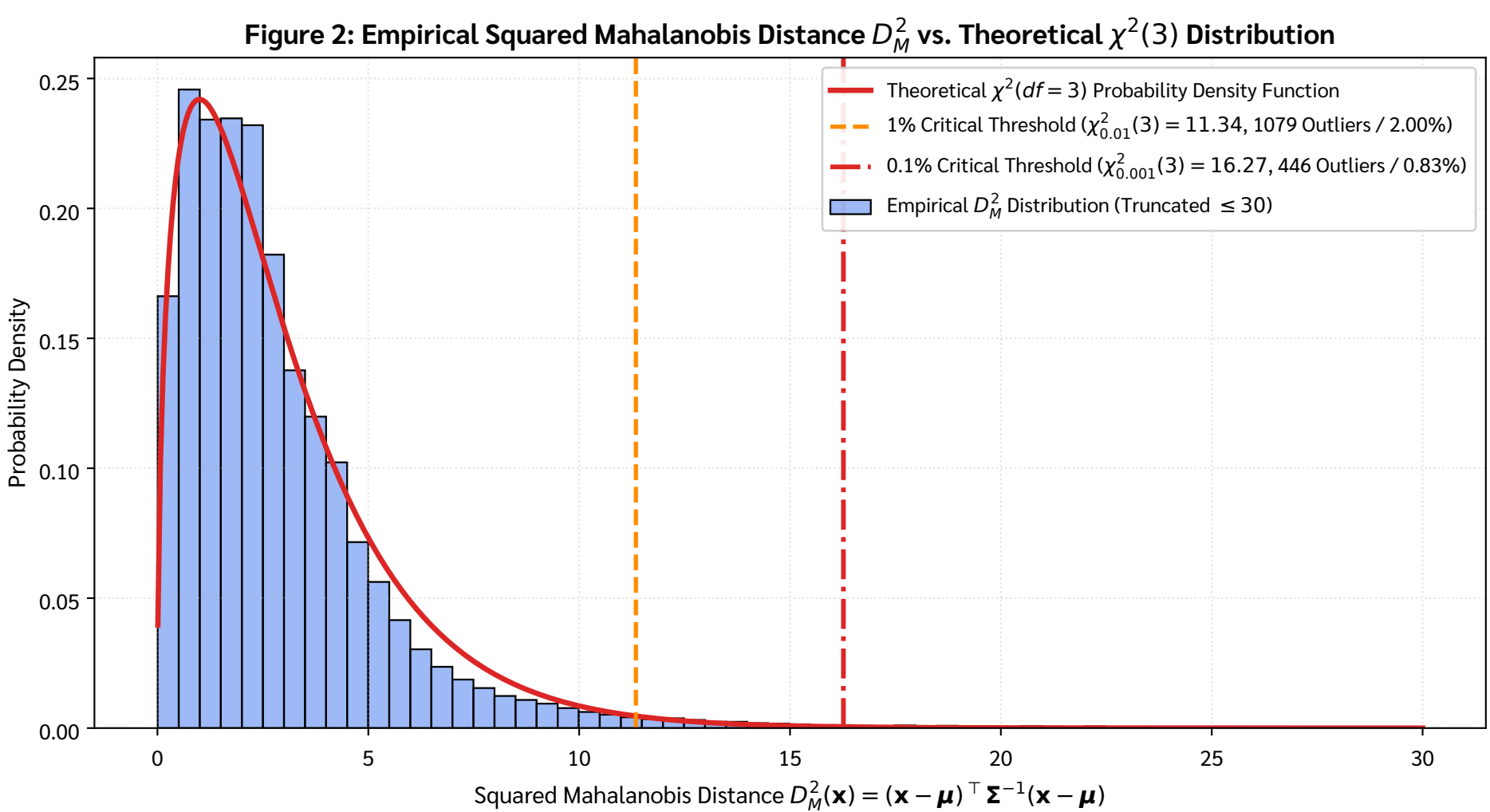
Figure 2: Empirical Squared Mahalanobis Distance D_M^2 vs. Theoretical $\chi^2(3)$ Distribution
Theoretical $\chi^2(df = 3)$ Probability Density Function
1% Critical Threshold ($\chi^2_{0.01}(3) = 11.34$, 1079 Outliers / 2.00%)
0.1% Critical Threshold ($\chi^2_{0.001}(3) = 16.27$, 446 Outliers / 0.83%)
Empirical D_M^2 Distribution (Truncated ≤ 30)
Probability Density
0.25
0.20
0.15
0.10
0.05
0.00
0
5
10
15
20
25
30
Squared Mahalanobis Distance $D_M^2(\mathbf{x}) = (\mathbf{x} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})$